

Roll No.

TPE-712

**B. TECH. (PE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

RESERVOIR MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define Reservoir Management. What are the various disciplines involved in Reservoir Management ? What is the difference between Reservoir Engineering and Reservoir Management ? (CO1)
- (b) How does technology play a vital role for reservoir management ? What are the various technology required for Reservoir characterization ? Explain in detail. (CO1)
- (c) What are the various techniques required for reservoir mapping and monitoring ? Explain each in detail. (CO1)

P. T. O.

2. (a) What are the five important different phases of oil and gas field ? Explain each in detail. Draw a cash flow diagram of an oil and gas field with time. (CO2)
- (b) What are three different phases of development plan in view of recovery mechanism ? What are the basic conditions for employing secondary recovery mechanism ? (CO2)
- (c) Design a Field Development Plan (FOP) with long-term production forecasts to support the business strategy and goals. (CO2)
3. (a) Explain reservoir modeling. Design a suitable workflow for making a reservoir model and explain its various steps in details. (CO3)
- (b) Explain Facies. Design a suitable workflow for facies modeling and explain its various steps in details. What are the essential data required for facies modeling ? (CO3)
- (c) Explain Variogram with figure. How this is useful for petrophysical modeling process ? Write the formula for variance calculation. (CO3)
4. (a) Explain primary recovery of an oil and gas field. What are the essential parameters responsible for this recovery ? What are the different methods applied for achieving good primary recovery ? (CO4)
- (b) When secondary recovery of an oil and gas field applied ? What are the essential parameters responsible for this recovery ? What are the different methods applied for achieving optimum secondary recovery ? (CO4)
- (c) Explain EOR and when it is applied in the field. What is the essential requirement for EOR application ? What are the different methods applied for achieving maximum oil from tertiary recovery ? (CO4)

(3)

5. (a) What are the three different categories of Reserve ? Explain each in detail in view of P90, P50 & P10. What are the deferent methods for reserve estimation ? (CO5)

(b) From logs, it has been determined that a productive formation has a bed thickness of 20 ft, a porosity of 25%, and a water saturation of 30%. The recovery factor is assumed to be 30%. The oil formation volume factor is 1.3. The areal extent of the reservoir is 200 acres.

Find : (CO5)

(i) HCPV expressed in ft

(ii) OIP expressed in bbl/acre

(iii) Reserves expressed in stock-tank barrels

(c) What is the difference between Risk and Uncertainty ? What are the various factor responsible for Uncertainty in measurement ? What are the SPE guidelines for E & P Companies for probability of Exceedance and non Exceedance for oil and gas assessment ? (CO5)

Roll No.

TPE-717

**B. TECH. (PE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

ADVANCED EOR TECHNIQUE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the classification of EOR as a thermal and non-thermal enhanced oil recovery. (CO1, CO2)
- (b) Explain the different oil production phases ? Why do we need FOR ? Currently available EOR is based on which parameter. (CO1, CO2)
- (c) Explain the different oil production phase. Why do we need EOR ? What are the different technique of FOR ? (CO3)
2. (a) Discuss the operational activities in pressure measurement during the production of crude oil. (CO3, CO4)

P. T. O.

- (b) Define static and bottom hole pressure. Write and define the different types oil field pressure calculation requirements. (CO1, CO4)
- (c) The data of depth and pressure is given in the below : (CO1, CO2)

Depth (m)	Press. (atm)
0	51
200	55.3
400	59.5
600	63.6
800	81.2
1000	99.0
1200	119.0
1400	141.0

Calculate the following :

- (i) Pressure gradients.
- (ii) Depth of gas- oil and gas water interface
- (iii) Pressures at GOC and WOC

Given : Depth of top perforation = 1140 m, GOC = 745 m,
WOC = 1160 m

3. (a) Define Arp's equation with the suitable diagram. What are the types of rate decline behavior ? Derive expressions for rate-time relationship for harmonic and exponential decline curve. (CO1)

(3)

- (b) Draw and explain with diagram of the effect of trapped gas on water flood recovery. Discuss the different selection patterns of water flooding. (CO1, CO2)
- (c) What are the typical displacement system in the miscible slug flooding? Explain with the illustration of schematic diagram. (CO2, CO5)
4. (a) Explain the ternary diagram. Discuss all the diagram by considering the different pressure and temperature along with the phase diagram inside the ternary diagram. (CO1, CO5)
- (b) What are the basic methods of miscible drive? Explain the phase conditions in the reservoir during high pressure gas injection. (CO4, CO5)
- (c) Explain the following : (CO3, CO6)
- (i) LPG slug injection
- (ii) Alcohol slug injection
5. (a) Discuss the reservoir selection and process design during advanced enhanced oil recovery. (CO4, CO6)
- (b) Draw the schematic diagram of MEOR and explain the each steps in increases the production rate of microorganisms during the enhanced oil recovery. (CO4, CO5)
- (c) Explain microbial recovery techniques and its principles, applications and potential. Discuss any projects for MEOR in Indian scenario. (CO2, CO3)

Roll No.

TPE-719

**B. TECH. (PE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

CITY GAS DISTRIBUTION

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are *twenty*.
(iv) Each sub-question carries 10 marks.
1. (a) Using Panhandle B equation, calculate the outlet pressure in a natural gas pipeline, with diameter 17.50 in, wall thickness 20 miles long. The gas flow rate is 145 MMSCFD at a flowing temperature of 65°F. The inlet pressure is 1000 psig and the gas gravity and viscosity are 0.6 and 0.00008 lb/ft-sec, respectively. Assume base pressure = 14.7 psia and base temperature = 70°F. Assume that the compressibility factor $Z = 0.75$ throughout and the pipeline efficiency is 0.95. (Show each step while answering). (CO1, CO2)

P. T. O.

- (b) Using Panhandle A equation, calculate base temperature for the following data given, $P_b = 14.7$ psia, $D = 13$ in, Pipe length = 0796 mi, gas Gravity = .71, $T = 75^\circ\text{F}$, $z = .919$, $p_1 = 570$ psia, $p_2 = 510$ psia, $q_h = 282$ MMSCFD. (Show each step while answering). (CO1, CO2)
- (c) Using Panhandle A equation, calculate Base Pressure for the following data given for a horizontal pipeline, $D = 12.09$ in, $L = 200$ mi, $e = 0.0006$ in, $T = 80^\circ\text{F}$, $Y_g = 0.70$, $T_b = 520^\circ\text{R}$, $p_1 = 600$ psia, $p_2 = 200$ psia, $z = .9188$, $q_h = 1076035$ cfh. (Show each step while answering).

(CO1, CO2)

2. (a) Using Weymouth equation, calculate the natural gas flow rate for gas and pipeline with the given information $T_b = 60^\circ\text{F}$, $P_b = 14.7$ psia, $D = 15$ in, Pipe length = 947 mi, gas Gravity = .68, $T = 80^\circ\text{F}$, $z = .919$, $p_1 = 510$ psia, $p_2 = 480$ psia. (Show each step while answering).

(CO1, CO2)

- (b) A section of a pipeline system is to handle 200 MMSCFD gas flow. The pipeline inlet pressure is 900 psig and the gas flows to a dehydrator operating at 800 psig. The allowable working pressure of the pipeline is 1480 psi. The following data are given : STC 14.73 psi, 60°F , average temperature 80°F , compressibility 0.67, specific gravity of gas 0.85, viscosity 0.01 cp, length 5 miles. Determine D , using Panhandle B equation. (Show each step while answering).

(CO1, CO2)

- (c) Define what is City Gas Distribution and also describe Evolution of City Gas Distribution in India with name of 10 Prominent CGD Companies working in India. (CO1, CO2)
3. (a) Define Gas metering and write all types of gas meters and explain any two gas meters. (CO3)
- (b) Explain Distribution and Transmission Pipeline and also write classification of pipeline based on pressure gas to be transported. (CO3)
- (c) Classify and explain Pipeline based on : (CO3)
- (i) Function
 - (ii) Access
 - (iii) Jurisdiction
4. (a) Differentiate between Steel pipelines and Polyethylene pipelines. (CO2, CO4)
- (b) Write different safety guidelines for gas leakage. (CO2, CO4)
- (c) Differentiate between CNG, PNG, LPG and write its advantages and disadvantages. (CO2, CO4)
5. (a) Explain what is gas leakage and describe how we can detect gas leakage. (CO5, CO6)
- (b) Write effects of gas leakage on environment and what different safety measures should be taken to avoid gas leakage. (CO5, CO6)

(c) For the following given data for a horizontal pipeline calculate/predict gas flow rate in ft^3/hr through the pipeline : (CO5, CO6)

(i) $D = 12.09 \text{ in}$

(ii) $L = 200 \text{ mi}$

(iii) $e = 0.006 \text{ in}$

(iv) $T = 80^\circ\text{F}$

(v) $Y_g = .70$

(vi) $T_b = 520^\circ\text{R}$

(vii) $p_b = 14.7 \text{ psia}$

(viii) $p_1 = 600 \text{ psia}$

(ix) $p_2 = 300 \text{ psia}$

Roll No.

TPE-724

**B. TECH. (PE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022
PETROLEUM ECONOMICS, RISK AND
UNCERTAINTY ANALYSIS**

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the objectives of petroleum economic and risk analysis ? Discuss why the analysis of infilling drilling is important for making oil and gas project more profitable and minimizing the risk. (CO1)
- (b) What is exploration license and relinquishment ? List and discuss the salient features of New Exploration License Policy (NELP). (CO1)
- (c) What are the phases of upstream oil and gas industry lifecycle ? Discuss. (CO1)
2. (a) What is cash flow statement ? What are the objectives of cash flow statement ? Discuss how the cash flow statement benefits the oil and gas industry projects ? (CO2)

P. T. O.

(2)

- (b) Describe "Indirect" method of ascertaining cash flow from operating activities. (CO2)
- (c) What is CAPEX and OPEX ? Explain, how to calculate capital expenditures (OPEX) for project. (CO2)
3. (a) What is Net Present Value ? Explain how the net present value helps in economic decision making in oil and gas industry. (CO3)
- (b) What is internal rate of return ? How is internal rate of return used in economic decision making ? (CO3)
- (c) What is meant by Capital Productivity Index in oil and gas industry ? Discuss the importance of capital productivity index in an organization. (CO3)
4. (a) What are the elements of fiscal regimes in oil and gas industry ? Discuss. (CO4)
- (b) What are revenue sharing contracts ? Differentiate between production sharing agreements and revenue sharing agreements in oil exploration. Why is the government shifting to the latter model in India ? (CO4)
- (c) Why the analysis of risk above the ground in oil and gas industry is important ? Discuss. (CO6)
5. (a) What is meant by risk and uncertainty in oil and gas industry ? Describe different types of risk ? What are the importance of risk analysis ? (CO5)
- (b) What are the geological risk in oil and gas exploration and production ? Discuss. (CO6)
- (c) What is meant by decision analysis ? Discuss the influencing factors of decision making in oil and gas industry. (CO5)